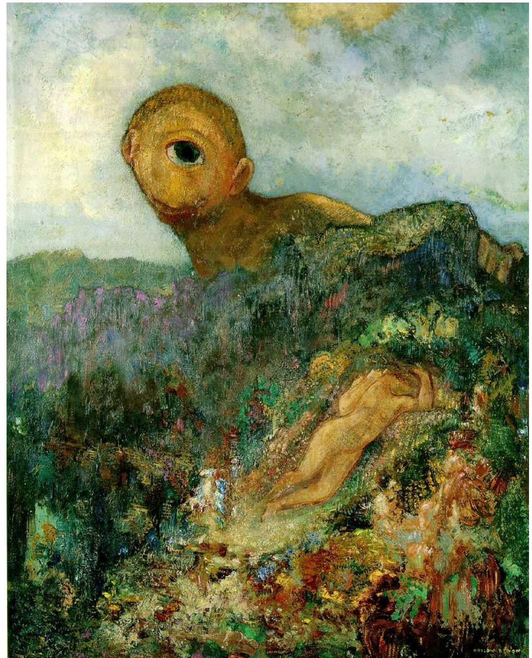
Geometry of a Single Camera: Calibration



Odilon Redon, *Cyclops*, 1914

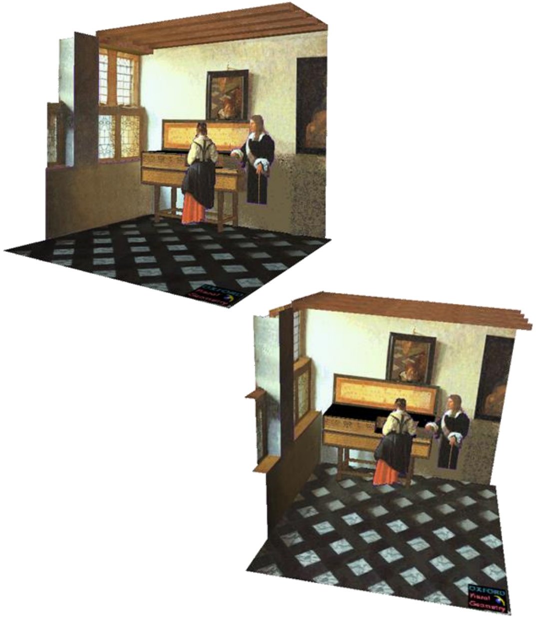
Slides: Svetlana Lazebnik

Outline

- Motivation
- Perspective projection model: review
- Intrinsic camera parameters → inside the cam parameters
- Extrinsic camera parameters → outside " " "
- Calibration
- Triangulation

Given an Image, Can We Recover 3D Structure?

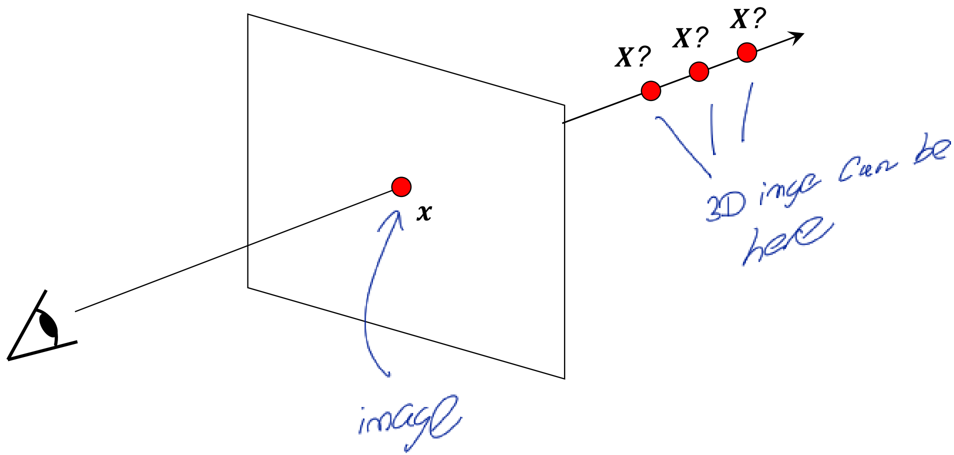
J. Vermeer, *Music Lesson*, 1662



A. Criminisi, M. Kemp, and A. Zisserman, [Bringing Pictorial Space to Life: computer techniques for the analysis of paintings](), *Proc. Computers and the History of Art*, 2002

Things Aren't Always as They Appear…

Single-View Ambiguity



Single-View Ambiguity

[Rashad Alakbarov shadow sculptures]

Anamorphic Perspective

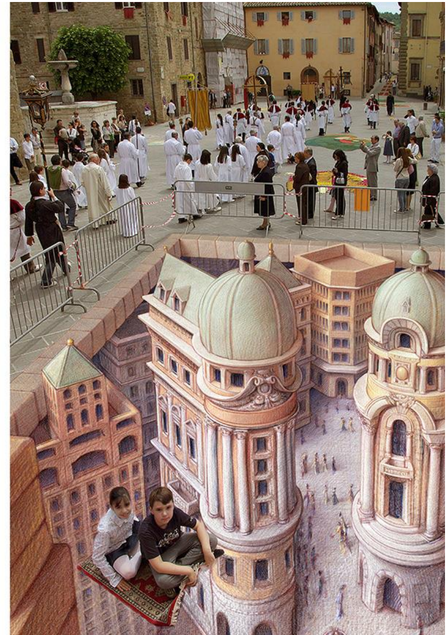


Image source

Anamorphic Perspective

H. Holbein The Younger, *The Ambassadors*, 1533

https://en.wikipedia.org/wiki/Anamorphosis

Our Goal: Recovery of 3D Structure

- When certain assumptions hold, we can recover structure from a single view

- In general, we need *multi-view geometry*

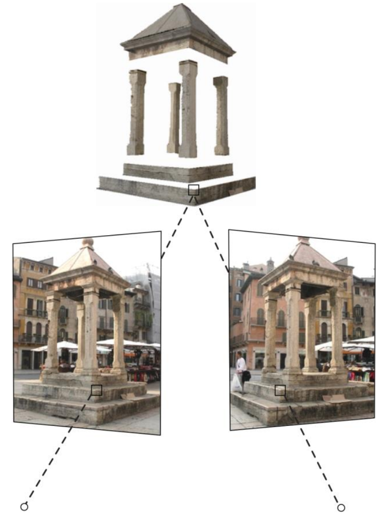


Image source

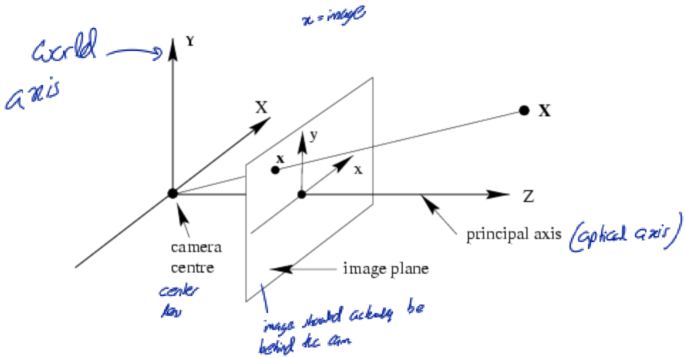
- But first, we need to understand the geometry of a single camera…

Outline

- Motivation
- Perspective projection model: review
- Intrinsic camera parameters
- Extrinsic camera parameters

Perspective Projection Model

Camera is same as lens
Image plane should be behind lens but due to similar triangles we can put it in front



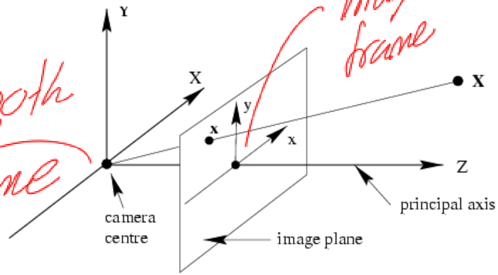
Normalized (camera) coordinate system: camera center is at the origin, the *principal axis* is the z-axis, x and y axes of the image plane are parallel to x and y axes of the world

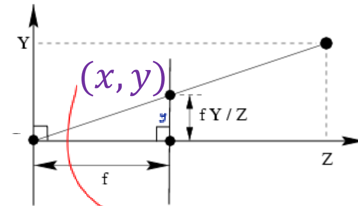
Perspective Projection Model

we consider image is created
at focal length (f)
There is no z to image (flat)

here both
cam and
world frame
same

image
frame





Cartesian

homogeneous

$$x = f\frac{X}{Z}, y = f\frac{Y}{Z}$$

$$\lambda\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

$\boldsymbol{x}$
Homogeneous coord. vec. of image point

$\boldsymbol{P}$
Camera projection matrix

$\boldsymbol{X}$
Homogeneous coord. vec. of 3D point

$$\lambda\boldsymbol{x} = \boldsymbol{P}\boldsymbol{X}$$

$$\boldsymbol{x} \equiv \boldsymbol{P}\boldsymbol{X}$$

Equality up to scale

3D

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

2D

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} \lambda x \\ \lambda y \\ \lambda \end{bmatrix}$$

to get cartesian divide by last

$$\frac{\lambda x}{\lambda} \qquad \frac{\lambda y}{\lambda}$$

This is why we add 1 at the end

---

step 01   Cartesian $(x, y) = \left(\frac{fX}{Z}, \frac{fY}{Z}\right)$ for image

homogeneous $\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \underbrace{\begin{pmatrix} fX \\ fY \\ Z \end{pmatrix}}_{x}$

step 02

$$\lambda x = K[I|0]\,X$$

$$\lambda \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \left[\begin{array}{ccc:c} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

$$\underbrace{\qquad\qquad}_{K}$$

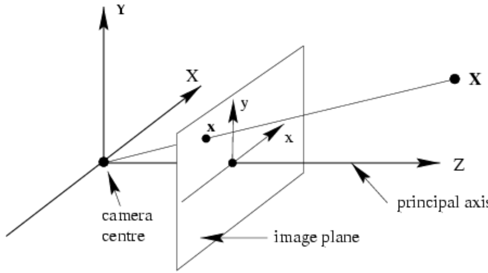
image wrt cam frame

object wrt world (cam to be exact) frame

Here in camera frame also we can measure 3D but we are measuring image plane so only 2D coords are taken.

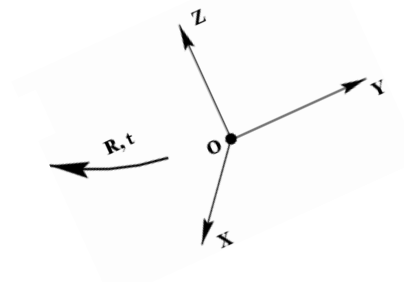
Camera Calibration: Overview

Now world and cam frame is different



world coordinate system

why? for multiple cameras.



Camera calibration: figuring out transformation from *world* coordinate system to *image* coordinate system

2D point (3x1)	=	Camera to pixel coord. trans. matrix K (3x3)	*Canonical* projection matrix $[I \mid 0]$ (3x4)	World to camera coord. trans. matrix $\begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix}$ (4x4)	3D point (4x1)
λx		***Intrinsic* camera parameters**: principal point, scaling factors		***Extrinsic* camera parameters**: rotation, translation	X

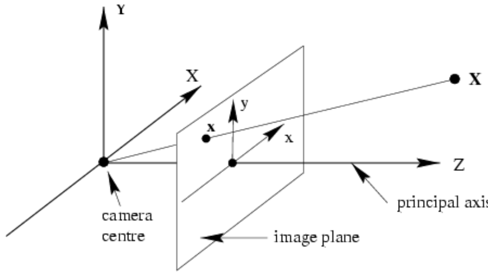
3x3 col of 0

eg focal length

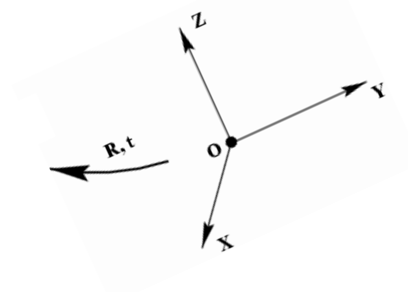
Rotational matrix 3x3 (orthonormal)

translation vector 3x1

Camera Calibration: Overview



world coordinate system



Camera calibration: figuring out transformation from *world* coordinate system to *image* coordinate system

2D point (3x1)	=	Camera to pixel coord. trans. matrix $\boldsymbol{K}$ (3x3)	*Canonical* projection matrix $[\boldsymbol{I} \mid \boldsymbol{0}]$ (3x4)	World to camera coord. trans. matrix $\begin{bmatrix} \boldsymbol{R} & \boldsymbol{t} \\ \boldsymbol{0}^T & 1 \end{bmatrix}$ (4x4)	3D point (4x1)
$\lambda \boldsymbol{x}$			$\boldsymbol{P} = \boldsymbol{K}[\boldsymbol{R} \mid \boldsymbol{t}]$		$\boldsymbol{X}$

$$\boldsymbol{P} = \boldsymbol{K}[\boldsymbol{R}|\boldsymbol{t}]$$

General camera projection matrix

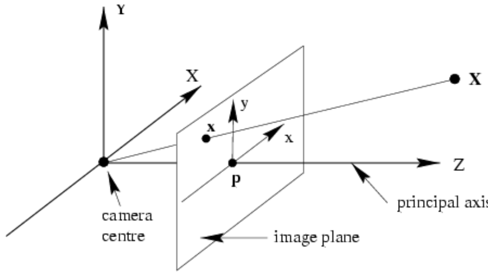
when world frame ≠ cam frame

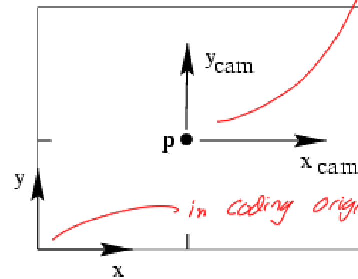
$$\underset{\substack{\uparrow \\ \text{image} \\ \text{wrt cam}}}{\lambda x} = K[I|0] \underbrace{\begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}}_{\text{becomes wrt cam}} \underset{\substack{\uparrow \\ \text{world}}}{X}$$

$$\lambda x = K[R|t]X$$

Intrinsic Parameters: Principal Point

earlier this way image origin



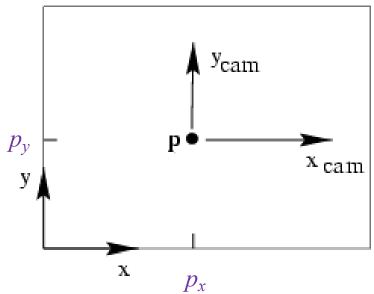


in coding origin of image will be in a corner

in here we take center = 0,0

- **Principal point (p):** point where principal axis intersects the image plane
- In the *normalized* coordinate system, the origin of the image is at the principal point
- In the *image* coordinate system: the origin is in the corner

Intrinsic Parameters: Principal Point



We want the principal point to map to (p_x, p_y) instead of $(0,0)$

$$x = f\frac{X}{Z} + p_x, \qquad y = f\frac{Y}{Z} + p_y$$

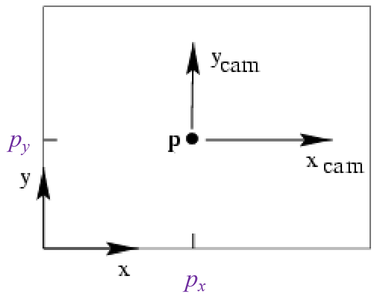
$$\lambda \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} fX + Zp_x \\ fY + Zp_y \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

only K will change

This is still in mm

not pixels.

Intrinsic Parameters: Principal Point



in actual calibration
p_c might not be exact center

Principal point: (p_x, p_y)

$$\begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

calibration matrix $\boldsymbol{K}$

Canonical projection matrix $[\boldsymbol{I} \mid \boldsymbol{0}]$

$\boldsymbol{P} = \boldsymbol{K}[\boldsymbol{I}|\boldsymbol{0}]$

Intrinsic Parameters: Principal Point

- What are the units of the focal length f and principal point coordinates (p_x, p_y)?
  - Same as world units – presumably metric units
- What units do we want for measuring image coordinates?
  - Pixel units
- Thus, we need to introduce scaling factors for mapping from world to pixel units

$$\begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

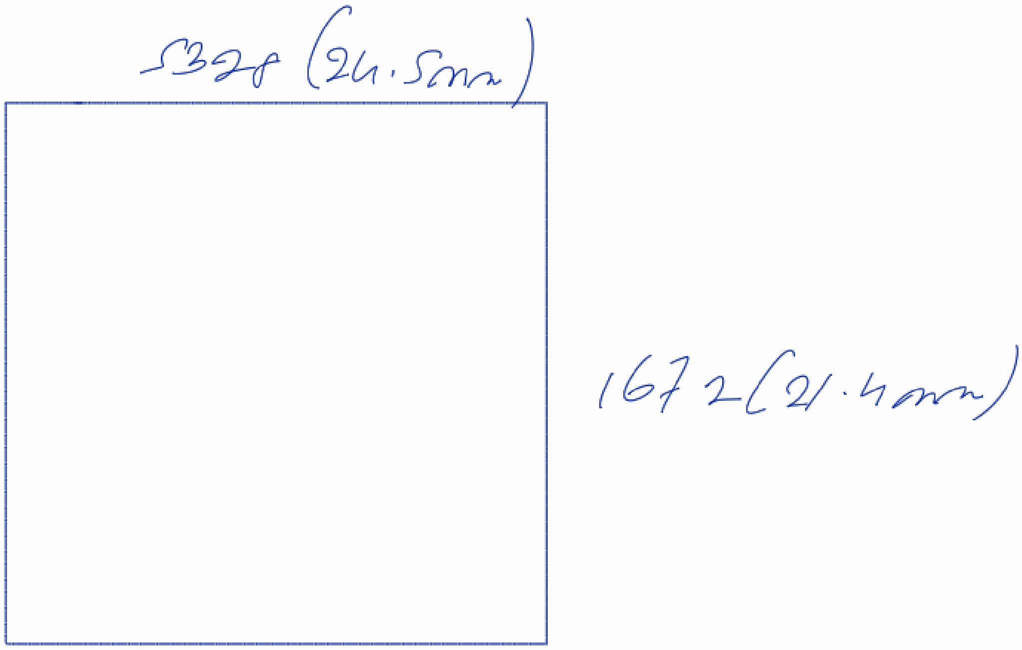
calibration matrix $\boldsymbol{K}$

Canonical projection matrix $[\boldsymbol{I} \mid \boldsymbol{0}]$

$\boldsymbol{P} = \boldsymbol{K}[\boldsymbol{I}|\boldsymbol{0}]$

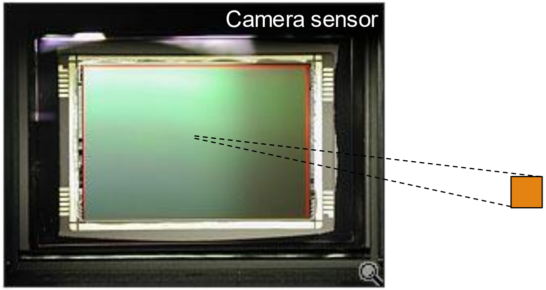
scaling factors depend on CCD

Cmos



should be able to find resolution on m_x, m_y without these details when focal length, object size, distance to object is given

Intrinsic Parameters: Scaling Factors



m_x pixels/m in horizontal direction,
m_y pixels/m in vertical direction

Pixel size (m): $\frac{1}{m_x} \times \frac{1}{m_y}$

$$\underbrace{\begin{bmatrix} m_x & 0 & 0 \\ 0 & m_y & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{pixels/m}}^{\text{Scaling factors}} \underbrace{\begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix}}_{\text{m}}^{\text{Calibration matrix } \boldsymbol{K} \text{ in metric units}} = \boxed{\underbrace{\begin{bmatrix} \alpha_x & 0 & \beta_x \\ 0 & \alpha_y & \beta_y \\ 0 & 0 & 1 \end{bmatrix}}_{\text{pixels}}^{\text{Calibration matrix } \boldsymbol{K} \text{ in pixel units}}}$$

K was updated twice

$$K \longrightarrow K \text{ with } p \text{ shift} \longrightarrow \overbrace{\begin{bmatrix} m_x & 0 & 0 \\ 0 & m_y & 0 \\ 0 & 0 & 1 \end{bmatrix} K_{old}}^{K_{new}}$$

$$\alpha_x = m_x f$$
$$\alpha_y = m_y f$$
$$\beta_x = m_x p_x$$
$$\beta_y = m_y p_y$$

K evolution

$$\begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} \alpha_x & 0 & \beta_x \\ 0 & \alpha_y & \beta_y \\ 0 & 0 & 1 \end{bmatrix}$$

Summary

$$\lambda x = K[R|t] X$$

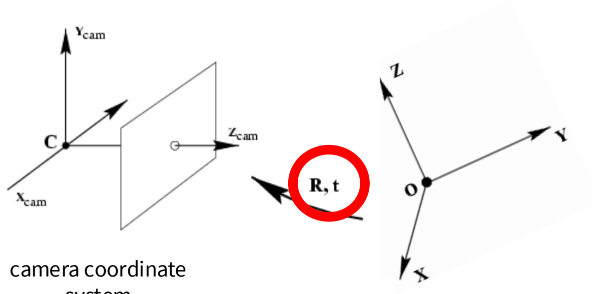
or if cam = world frame

$$\lambda x = K[I|0] X$$

where $K = \begin{bmatrix} m_x & 0 & 0 \\ 0 & m_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix}$

cases in pixels

Extrinsic Parameters: Rotation and Translation



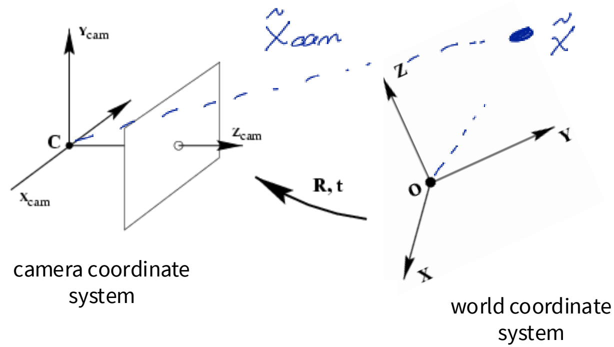
camera coordinate system

world coordinate system

In general, the camera coordinate frame will be related to the world coordinate frame by a rotation and a translation

N ⇒ only 3 numbers not 4

Extrinsic Parameters: Rotation and Translation



In general, the camera coordinate frame will be related to the world coordinate frame by a rotation and a translation

In non-homogeneous coordinates, the transformation from world to normalized camera coordinate system is given by:

$$\tilde{X}_{cam} = R(\tilde{X} - \tilde{C}) = R\tilde{X} + t$$

coords. of point in normalized camera frame

3x3 rotation matrix

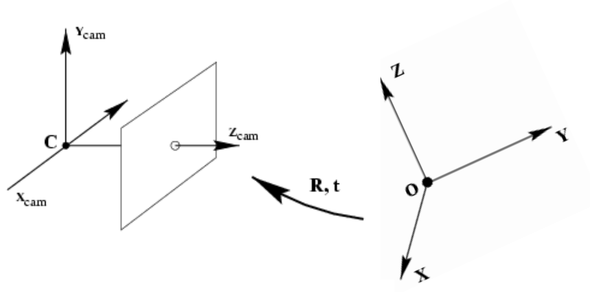
coords. of a point in world frame

coords. of camera center in world frame

Extrinsic Parameters: Rotation and Translation

not new

$$\lambda x = K[I|0]\underbrace{\begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} X_{world}}_{X_{cam}}$$



In *non-homogeneous* coordinates:

$$\widetilde{X}_{\text{cam}} = R\widetilde{X} + t$$

In *homogeneous* coordinates:

$$\underset{4\times 1}{\begin{pmatrix} \widetilde{X}_{\text{cam}} \\ 1 \end{pmatrix}} = \underset{4\times 4}{\begin{bmatrix} \underset{3\times 3}{R} & \underset{3\times 1}{t} \\ \underset{1\times 3}{0^T} & \underset{1\times 1}{1} \end{bmatrix}} \underset{4\times 1}{\begin{pmatrix} \underset{3\times 1}{\widetilde{X}} \\ \underset{1\times 1}{1} \end{pmatrix}}$$

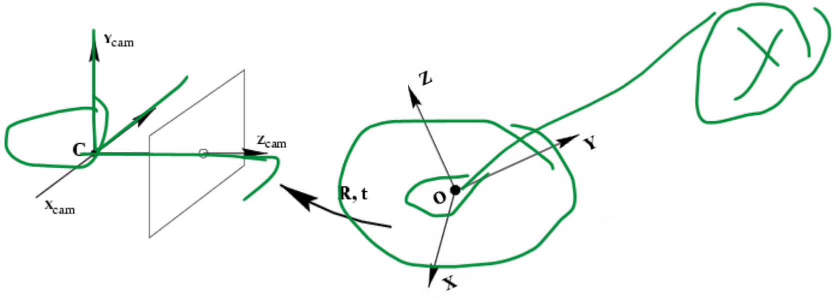
to make homogeneous

3D transformation matrix (4 x 4)

t is multiplied with this so 1 is important

now we have represented x in the cam coord frame X_{cam}

Extrinsic Parameters: Rotation and Translation



In *non-homogeneous* coordinates:

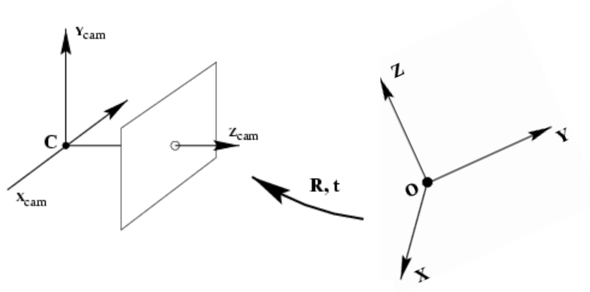
$$\tilde{\boldsymbol{X}}_{\text{cam}} = \boldsymbol{R}\tilde{\boldsymbol{X}} + \boldsymbol{t}$$

In *homogeneous* coordinates:

$$\boldsymbol{X}_{\text{cam}} = \begin{bmatrix} \boldsymbol{R} & \boldsymbol{t} \\ \boldsymbol{0}^{T} & 1 \end{bmatrix} \boldsymbol{X}$$

3D transformation matrix (4 x 4)

Extrinsic Parameters: Rotation and Translation



In *non-homogeneous* coordinates:

$$\widetilde{\boldsymbol{X}}_{\text{cam}} = \boldsymbol{R}\widetilde{\boldsymbol{X}} + \boldsymbol{t}$$

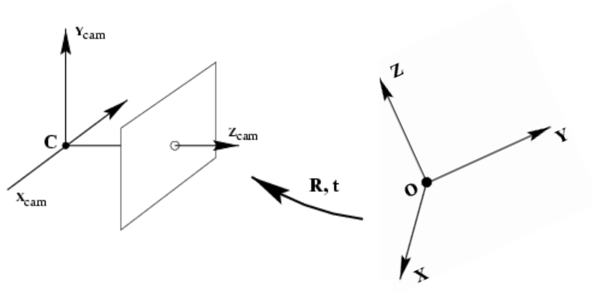
In *homogeneous* coordinates:

$$\boldsymbol{X}_{\text{cam}} = \begin{bmatrix} \boldsymbol{R} & \boldsymbol{t} \\ \boldsymbol{0}^T & 1 \end{bmatrix} \boldsymbol{X}$$

Transformation from normalized 3D coordinates to pixel image coordinates:

$$\lambda \boldsymbol{x} = \boldsymbol{K}[\boldsymbol{I}|\boldsymbol{0}]\boldsymbol{X}_{\text{cam}}$$

Extrinsic Parameters: Rotation and Translation



$$\underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_{\text{this is just used to lob the last row}} \begin{bmatrix} R_{3\times 3} & \begin{matrix} t_x \\ t_y \\ t_z \end{matrix} \\ 0\,0\,0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$K[I|0]$ is intrinsic (cam params)

$\begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}$ is extrinsic (world part)

In *homogeneous* coordinates:

$$\lambda \boldsymbol{x} = \underset{3\times 3}{\boldsymbol{K}}[\boldsymbol{I}|\boldsymbol{0}]\begin{bmatrix} \boldsymbol{R} & \boldsymbol{t} \\ \boldsymbol{0}^T & 1 \end{bmatrix}\boldsymbol{X}$$

Finally:

$$\underline{\lambda \boldsymbol{x} = \boldsymbol{K}[\boldsymbol{R}|\boldsymbol{t}]\boldsymbol{X}} \qquad \boldsymbol{t} = -\boldsymbol{R}\widetilde{\boldsymbol{C}}$$

convert m to pixel

$$P^c = H_w^c P^w$$

$$\tilde{X}_{cam} = H_{world}^{cam} \tilde{X}_{world}$$

So R is the rot matrix of world wrt camera

so $$H_{world}^{cam} = \begin{bmatrix} R_{world}^{cam} & t_{world}^{cam} \\ 1 & 0 \end{bmatrix}$$

$$t_{world}^{cam} = -[R_c^w]^T t_{cam}^{world}$$

$$= -R_w^c t_c^w$$

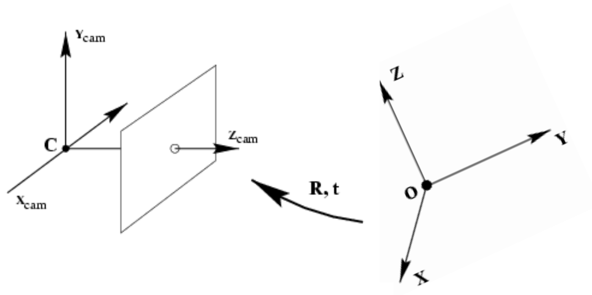
$$= -R\tilde{C}$$

$$\lambda x = PX$$

$$\downarrow$$

$$K[R|t]$$

Extrinsic Parameters: Rotation and Translation



$$\begin{bmatrix} \tilde{X}_{cam} \\ 1 \end{bmatrix} = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \tilde{X} \\ 1 \end{bmatrix}$$

$$\tilde{X}_{cam} = R\tilde{X} + t = R\tilde{X} - R\tilde{C} = R(\tilde{X} - \tilde{C})$$

$$\lambda x = \underline{K}[R|t]X \qquad t = -R\tilde{C}$$

coords. of camera
center
in world frame

- What is the projection of the camera center in world coordinates?

wrt world, cam
center is $\tilde{C}$
wrt cam center, $\tilde{C}$ is 0

$$PC = K[R \mid -R\tilde{C}]\begin{pmatrix} \tilde{C} \\ 1 \end{pmatrix} = K(R\tilde{C} - R\tilde{C}) = 0$$

- The camera center is the null space of the projection matrix!

Summary

$$\lambda x = K[I|0]\begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} X$$

$$= \underbrace{K[R|t]}_{P} X$$

$$K = \begin{bmatrix} \alpha_x & 0 & \beta_x \\ 0 & \alpha_y & \beta_y \\ 0 & 0 & 1 \end{bmatrix}$$ and

$$\boxed{R_w^c \quad t_w^c}$$

$$\tilde{X}_{cam} = R\tilde{X} + t$$

$$= R(\tilde{X} - \tilde{C})$$

wrt cam

wrt world

Find eqn

$$\lambda x = K[R|t] \; X$$

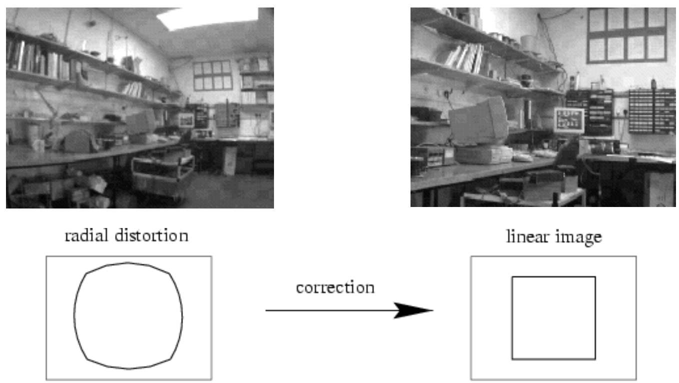
$$= K[R| -R\tilde{C}] X$$

Camera Parameters: Summary

$$\boldsymbol{P} = \boldsymbol{K}[\boldsymbol{R}|\boldsymbol{t}]$$

- Intrinsic parameters
  - Principal point coordinates
  - Focal length
  - Pixel magnification factors
  - *Skew (non-rectangular pixels) – not important in practice*
  - *Radial distortion – important in practice!*

$$\boldsymbol{K} = \begin{bmatrix} m_x & 0 & 0 \\ 0 & m_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & \beta_x \\ 0 & \alpha_y & \beta_y \\ 0 & 0 & 1 \end{bmatrix}$$



Camera Parameters: Summary

$$P = K[R|t]$$

- Intrinsic parameters
  - Principal point coordinates $p_x \; p_y$
  - Focal length f
  - Pixel magnification factors $\alpha_x \; \alpha_y \; \beta_x \; \beta_y$
  - *Skew (non-rectangular pixels) – not important in practice*
  - *Radial distortion – important in practice!*
- Extrinsic parameters
  - Rotation and translation relative to world coordinate system

Outline

- Motivation
- Perspective projection model: review
- Intrinsic camera parameters
- Extrinsic camera parameters
- Calibration

Camera Calibration

finding the matrix

$$\lambda \boldsymbol{x} = \boldsymbol{K}[\boldsymbol{R} \ \ \boldsymbol{t}]\boldsymbol{X}$$

3x1 3x3 3x4 4x1

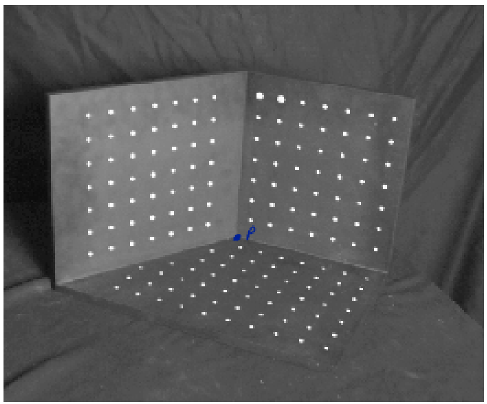
$$\begin{pmatrix} \lambda x \\ \lambda y \\ \lambda \end{pmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

finding these parameters is calibration.

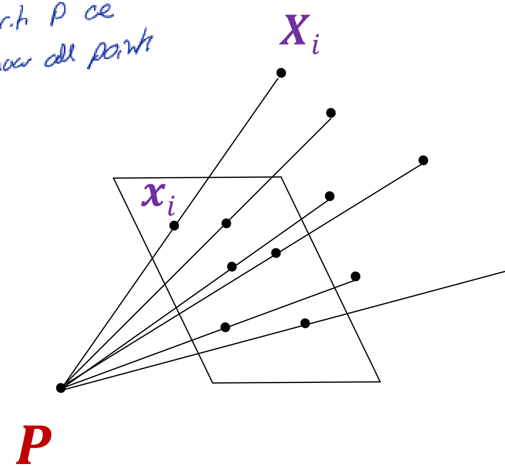
11 degrees of freedom

Camera Calibration

- Given n points with known 3D coordinates $\boldsymbol{X}_i$ and known image projections $\boldsymbol{x}_i$, estimate the camera parameters



w.r.t P we know all points



Camera Calibration

- Given n points with known 3D coordinates $\boldsymbol{X}_i$ and known image projections $\boldsymbol{x}_i$, estimate the camera parameters

Known 2D image coords		Known 3D locations		
x	y	X	Y	Z
880	214	312.747	309.140	30.086
43	203	305.796	311.649	30.356
270	197	307.694	312.358	30.418
886	347	310.149	307.186	29.298
745	302	311.937	310.105	29.216
943	128	311.202	307.572	30.682
476	590	307.106	306.876	28.660
419	214	309.317	312.490	30.230
317	335	307.435	310.151	29.318
783	521	308.253	306.300	28.881
235	427	306.650	309.301	28.905
665	429	308.069	306.831	29.189
655	362	309.671	308.834	29.029
427	333	308.255	309.955	29.267
412	415	307.546	308.613	28.963
746	351	311.036	309.206	28.913
434	415	307.518	308.175	29.069
525	234	309.950	311.262	29.990
716	308	312.160	310.772	29.080
602	187	311.988	312.709	30.514

Image credit: J. Hays

Camera Calibration: Linear Method

- Recall homography fitting:

To get rid of λ

row

$$\lambda \mathbf{x}_i = \mathbf{P}\mathbf{X}_i \qquad \mathbf{x}_i \times \mathbf{P}\mathbf{X}_i = \mathbf{0} \qquad \begin{pmatrix} x_i \\ y_i \\ 1 \end{pmatrix} \times \begin{pmatrix} \mathbf{P}_1^T \mathbf{X}_i \\ \mathbf{P}_2^T \mathbf{X}_i \\ \mathbf{P}_3^T \mathbf{X}_i \end{pmatrix} = \mathbf{0}$$

(3×1) (4×1) parallel

The cross product between 2 vectors can be written like this.

$$\begin{bmatrix} \mathbf{0}^T & -\mathbf{X}_i^T & y_i \mathbf{X}_i^T \\ \mathbf{X}_i^T & \mathbf{0}^T & -x_i \mathbf{X}_i^T \\ -y_i \mathbf{X}_i^T & x_i \mathbf{X}_i^T & \mathbf{0}^T \end{bmatrix} \begin{pmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \\ \mathbf{P}_3 \end{pmatrix} = 0$$

P = 0 trivial

One match gives two linearly independent constraints

Camera Calibration: Linear Method

- Final linear system:

$$\begin{bmatrix} \mathbf{0}^T & \boldsymbol{X}_1^T & -y_1\boldsymbol{X}_1^T \\ \boldsymbol{X}_1^T & \mathbf{0}^T & -x_1\boldsymbol{X}_1^T \\ \cdots & \cdots & \cdots \\ \mathbf{0}^T & \boldsymbol{X}_n^T & -y_n\boldsymbol{X}_n^T \\ \boldsymbol{X}_n^T & \mathbf{0}^T & -x_n\boldsymbol{X}_n^T \end{bmatrix} \begin{pmatrix} \boldsymbol{P}_1 \\ \boldsymbol{P}_2 \\ \boldsymbol{P}_3 \end{pmatrix} = 0 \qquad \boldsymbol{A}\boldsymbol{p} = 0$$

- One 2D/3D correspondence gives two linearly independent equations
  - The projection matrix has 11 degrees of freedom
  - 6 correspondences needed for a minimal solution
- Homogeneous least squares: find $\boldsymbol{p}$ minimizing $\|\boldsymbol{A}\boldsymbol{p}\|^2$
  - Solution is eigenvector of $\boldsymbol{A}^T\boldsymbol{A}$ corresponding to smallest eigenvalue

Similar to TLS

Camera Calibration: Linear Method

- Final linear system:

$$\begin{bmatrix} \mathbf{0}^T & \boldsymbol{X}_1^T & -y_1\boldsymbol{X}_1^T \\ \boldsymbol{X}_1^T & \mathbf{0}^T & -x_1\boldsymbol{X}_1^T \\ \dots & \dots & \dots \\ \mathbf{0}^T & \boldsymbol{X}_n^T & -y_n\boldsymbol{X}_n^T \\ \boldsymbol{X}_n^T & \mathbf{0}^T & -x_n\boldsymbol{X}_n^T \end{bmatrix} \begin{pmatrix} \boldsymbol{P}_1 \\ \boldsymbol{P}_2 \\ \boldsymbol{P}_3 \end{pmatrix} = 0 \qquad \boldsymbol{A}\boldsymbol{p} = 0$$

- What if all the n 3D points are *coplanar*, i.e., there exists a set of line parameters $\boldsymbol{\Pi}^T = (a, b, c, d)^T$ such that $\boldsymbol{\Pi}^T\boldsymbol{X}_i = 0$ for all i?
  - Then we will get *degenerate solutions* $(\boldsymbol{\Pi}, \mathbf{0}, \mathbf{0})$, $(\mathbf{0}, \boldsymbol{\Pi}, \mathbf{0})$, or $(\mathbf{0}, \mathbf{0}, \boldsymbol{\Pi})$

This is the linear method

Camera Calibration: Linear vs. Nonlinear

- Linear calibration is easy to formulate and solve, but it doesn't directly tell us the camera parameters

$$\begin{pmatrix} \lambda x \\ \lambda y \\ \lambda \end{pmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} \quad \text{vs.} \quad \lambda \boldsymbol{x} = \boldsymbol{K}[\boldsymbol{R} \mid \boldsymbol{t}]\boldsymbol{X}$$

In nonlinear method
we get an initial guess for p and calculate
$\hat{x}_i$ and compare with real x_i

Camera Calibration: Linear vs. Nonlinear

- Linear calibration is easy to formulate and solve, but it doesn't directly tell us the camera parameters
- In practice, *non-linear* methods are preferred
  - Write down objective function in terms of intrinsic and extrinsic parameters, as sum of squared distances between measured 2D points and estimated projections of 3D points:

$$\sum_i \|\underbrace{\mathrm{proj}(\overbrace{\boldsymbol{K}[\boldsymbol{R} \mid \boldsymbol{t}]\boldsymbol{X}_i}^{\hat{x}_i})}_{\text{projected coord.}} - \underbrace{\boldsymbol{x}_i}_{\text{image coords}}\|_2^2$$

need to minimize this by finding K R t
this is better than the previous method.
↓
linear

  - Can include radial distortion and constraints such as known focal length, orthogonality, visibility of points
  - Minimize error using non-linear optimization package
  - Can initialize solution with output of linear method (perform QR decomposition to get ***K*** and ***R*** from ***P***)

Outline

- Motivation
- Perspective projection model: review
- Intrinsic camera parameters
- Extrinsic camera parameters
- Calibration
- Triangulation

We try to find real location
from 2 images

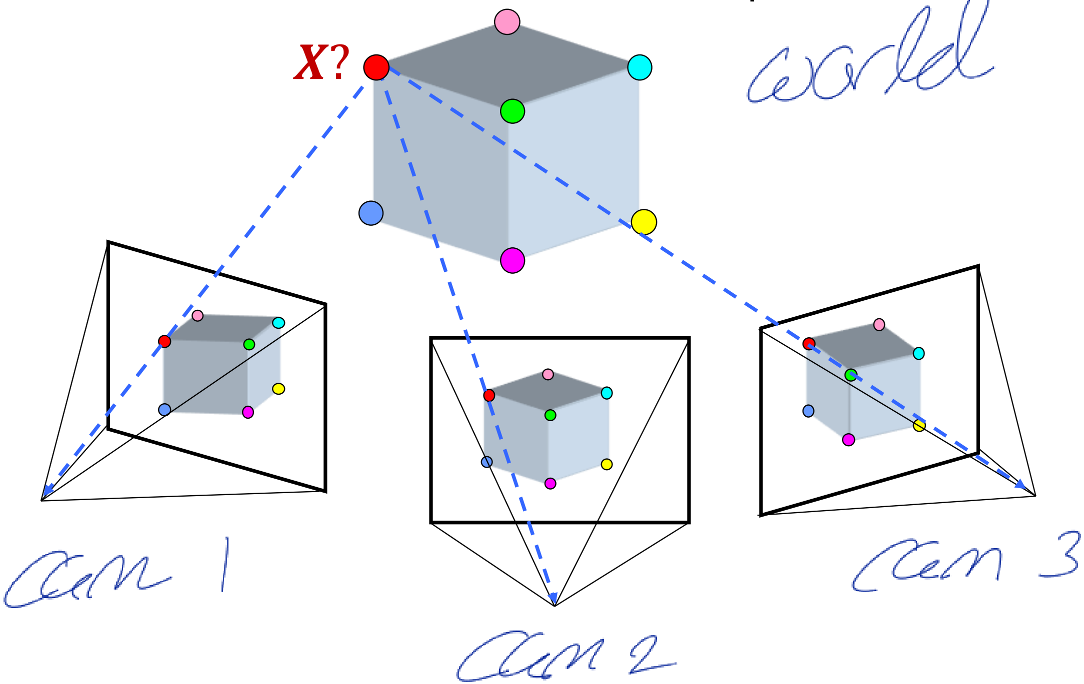
A Taste of Multi-View Geometry: Triangulation

Given projections of a 3D point in two or more images (with known camera matrices), find the coordinates of the point

- These are 3D triangulation i finding the 3D coord

A Taste of Multi-View Geometry: Triangulation

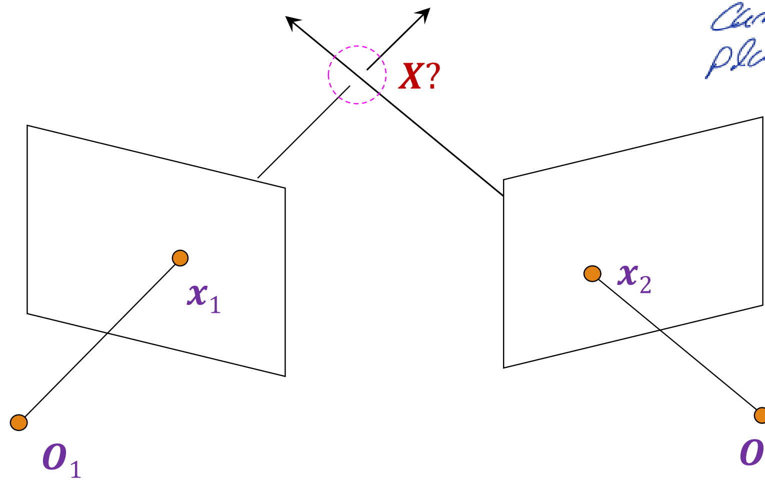
Given projections of a 3D point in two or more images (with known camera matrices), find the coordinates of the point



Triangulation

Given projections of a 3D point in two or more images (with known camera matrices), find the coordinates of the point

Now in reverse if we try to find world coord using cam coord in image plane.

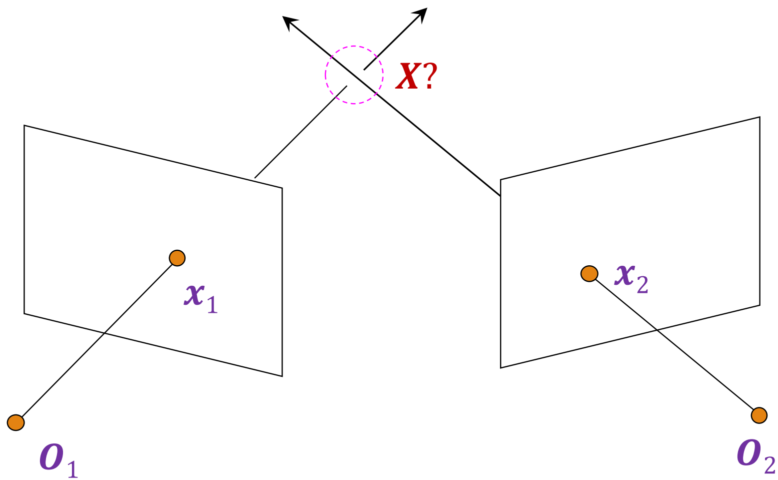


Camera center 1

Camera center 2

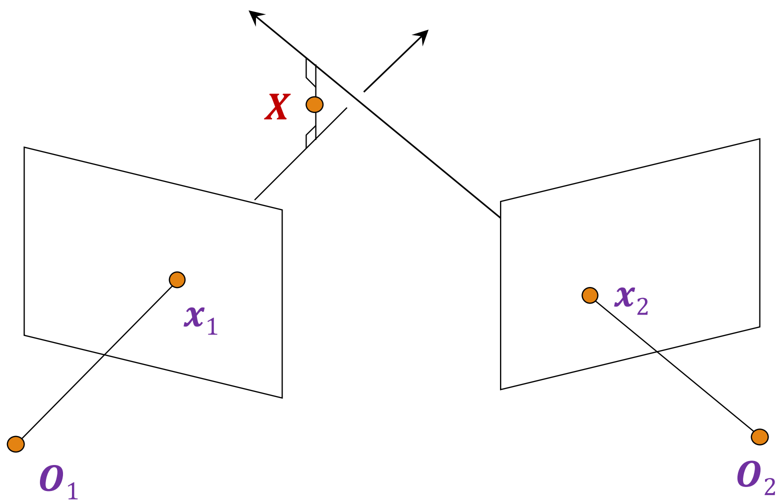
Triangulation

We want to intersect the two visual rays corresponding to $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, but because of noise and numerical errors, they don't meet exactly



Triangulation: Geometric approach

Find shortest segment connecting the two viewing rays and let $\boldsymbol{X}$ be the midpoint of that segment

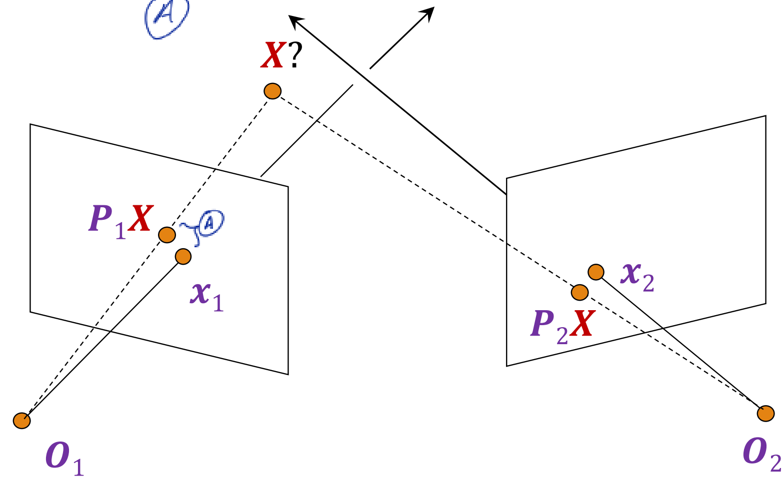


Triangulation: Nonlinear approach

Find $\boldsymbol{X}$ that minimizes

camera matrix

$$\underbrace{\|\mathrm{proj}(\boldsymbol{P}_1\boldsymbol{X}) - \boldsymbol{x}_1\|_2^2}_{A} + \|\mathrm{proj}(\boldsymbol{P}_2\boldsymbol{X}) - \boldsymbol{x}_2\|_2^2$$



find X that minimize the total error from all cameras

Triangulation: Linear approach

from 2 camera $\begin{cases} \lambda_1 \boldsymbol{x}_1 = \boldsymbol{P}_1 \boldsymbol{X} \\ \lambda_2 \boldsymbol{x}_2 = \boldsymbol{P}_2 \boldsymbol{X} \end{cases}$

$$\boldsymbol{x}_1 \times \boldsymbol{P}_1 \boldsymbol{X} = \boldsymbol{0} \qquad [\boldsymbol{x}_{1\times}]\boldsymbol{P}_1 \boldsymbol{X} = \boldsymbol{0}$$

$$\boldsymbol{x}_2 \times \boldsymbol{P}_2 \boldsymbol{X} = \boldsymbol{0} \qquad [\boldsymbol{x}_{2\times}]\boldsymbol{P}_2 \boldsymbol{X} = \boldsymbol{0}$$

Rewrite cross product as matrix multiplication:

$$\boldsymbol{a} \times \boldsymbol{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = [\boldsymbol{a}_\times]\boldsymbol{b}$$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \times \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Triangulation: Linear approach

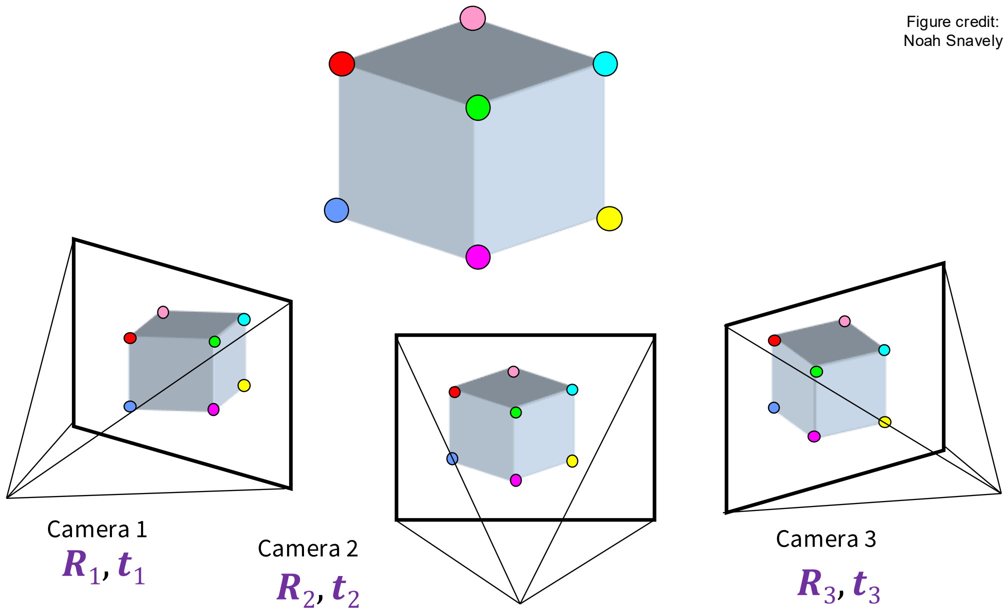
$$\lambda_1 \boldsymbol{x}_1 = \boldsymbol{P}_1 \boldsymbol{X} \qquad \boldsymbol{x}_1 \times \boldsymbol{P}_1 \boldsymbol{X} = \boldsymbol{0} \qquad [\boldsymbol{x}_{1\times}] \boldsymbol{P}_1 \boldsymbol{X} = \boldsymbol{0}$$

$$\lambda_2 \boldsymbol{x}_2 = \boldsymbol{P}_2 \boldsymbol{X} \qquad \boldsymbol{x}_2 \times \boldsymbol{P}_2 \boldsymbol{X} = \boldsymbol{0} \qquad [\boldsymbol{x}_{2\times}] \boldsymbol{P}_2 \boldsymbol{X} = \boldsymbol{0}$$

Two independent equations
each in terms of
three unknown entries of $\boldsymbol{X}$

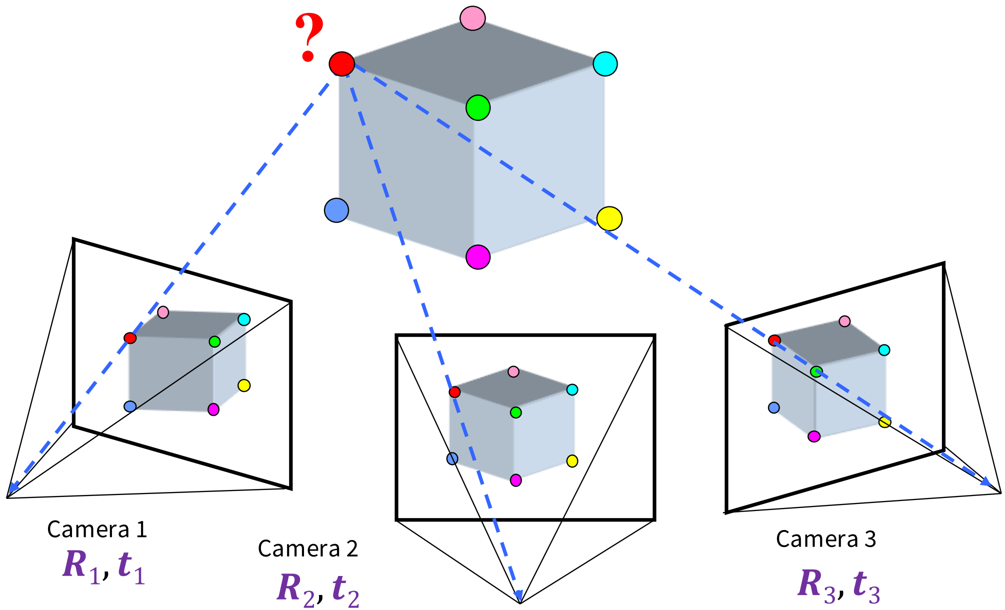
Preview: Structure from motion

Figure credit:
Noah Snavely



Given 2D point correspondences between multiple images, compute the camera parameters and the 3D points

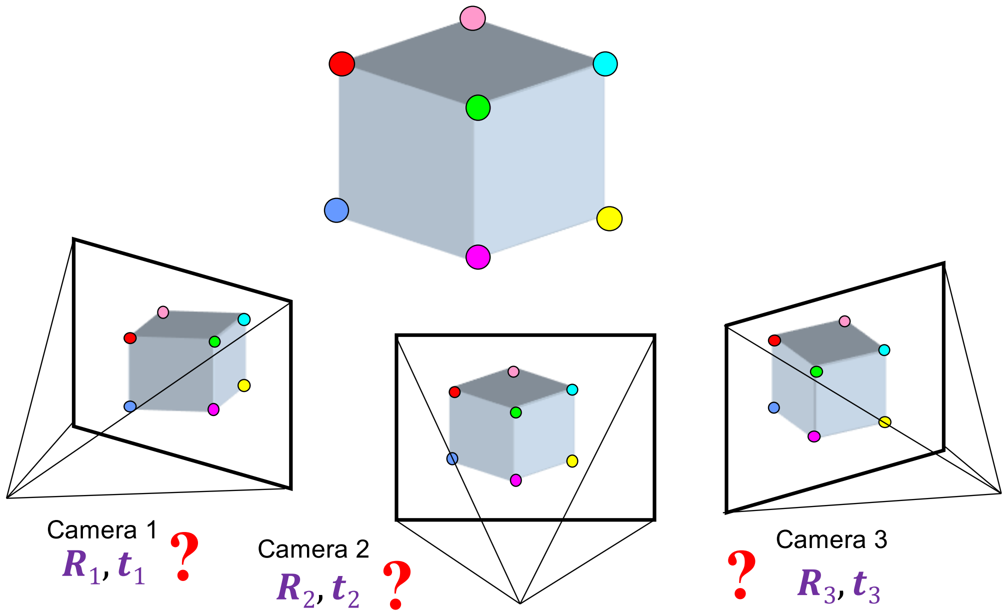
Preview: Structure from Motion



Structure: Given *known cameras* and projections of the same 3D point in two or more images, compute the 3D coordinates of that point

→ Triangulation!

Preview: Structure from Motion



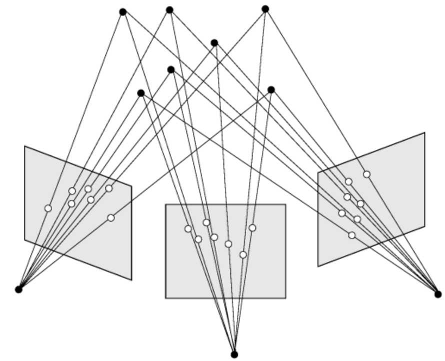
Motion: Given a set of *known* 3D points seen by a camera, compute the camera parameters

→Calibration!

Bundle Adjustment

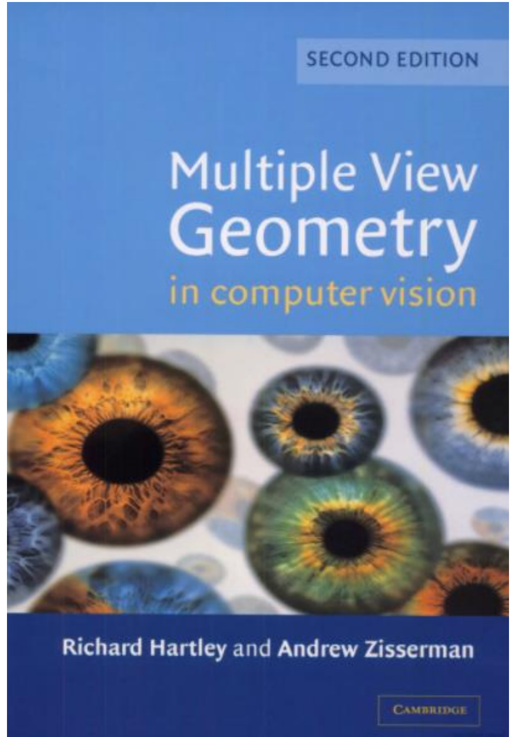
Simultaneous refining of the 3D coordinates describing the scene geometry, the parameters of th relative motion, and the optical characteristics of the camera(s) employed to acquire the images, given a set of images depicting several 3D points from different viewpoints.

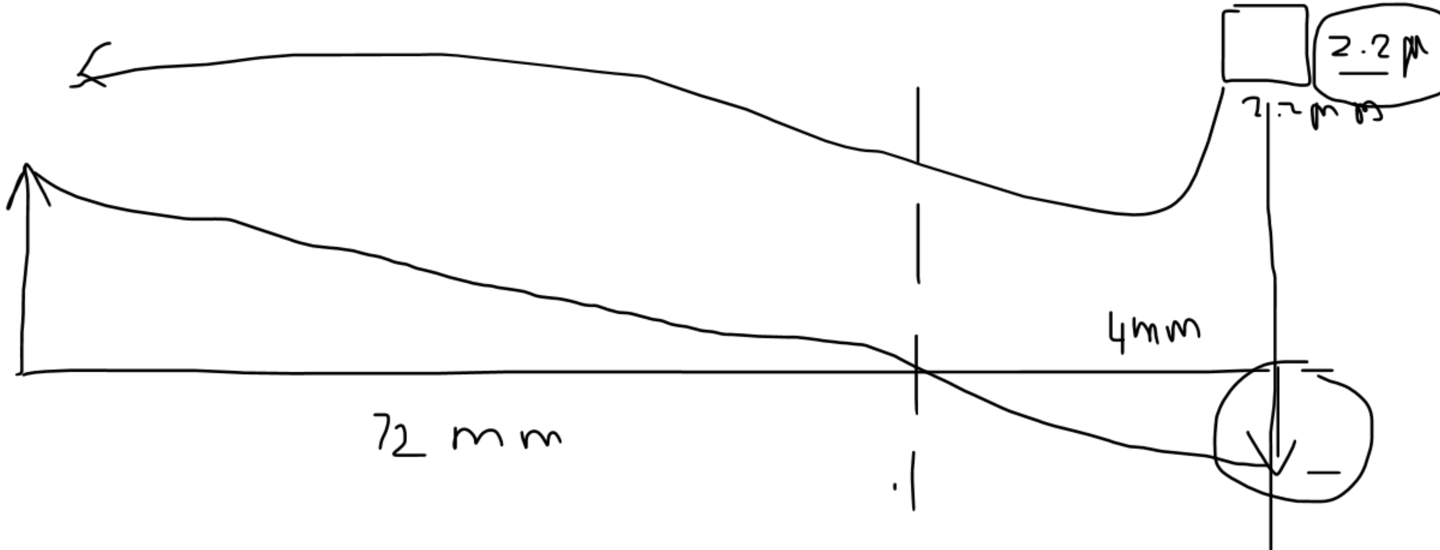
Its name refers to the geometrical bundles of light rays originating from each 3D feature and convergin on each camera's optical center, which are adjusted optimally according to an optimality criterion involving the corresponding image projections of all points.



https://en.wikipedia.org/wiki/Bundle_adjustment

Multi-View Geometry Book





$$x \rightarrow \frac{2.2 \times 10^{-3} \times \frac{72}{4}}{}$$

D

y z ✓

L